EE372 Design Showcase

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Group Members



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Agenda

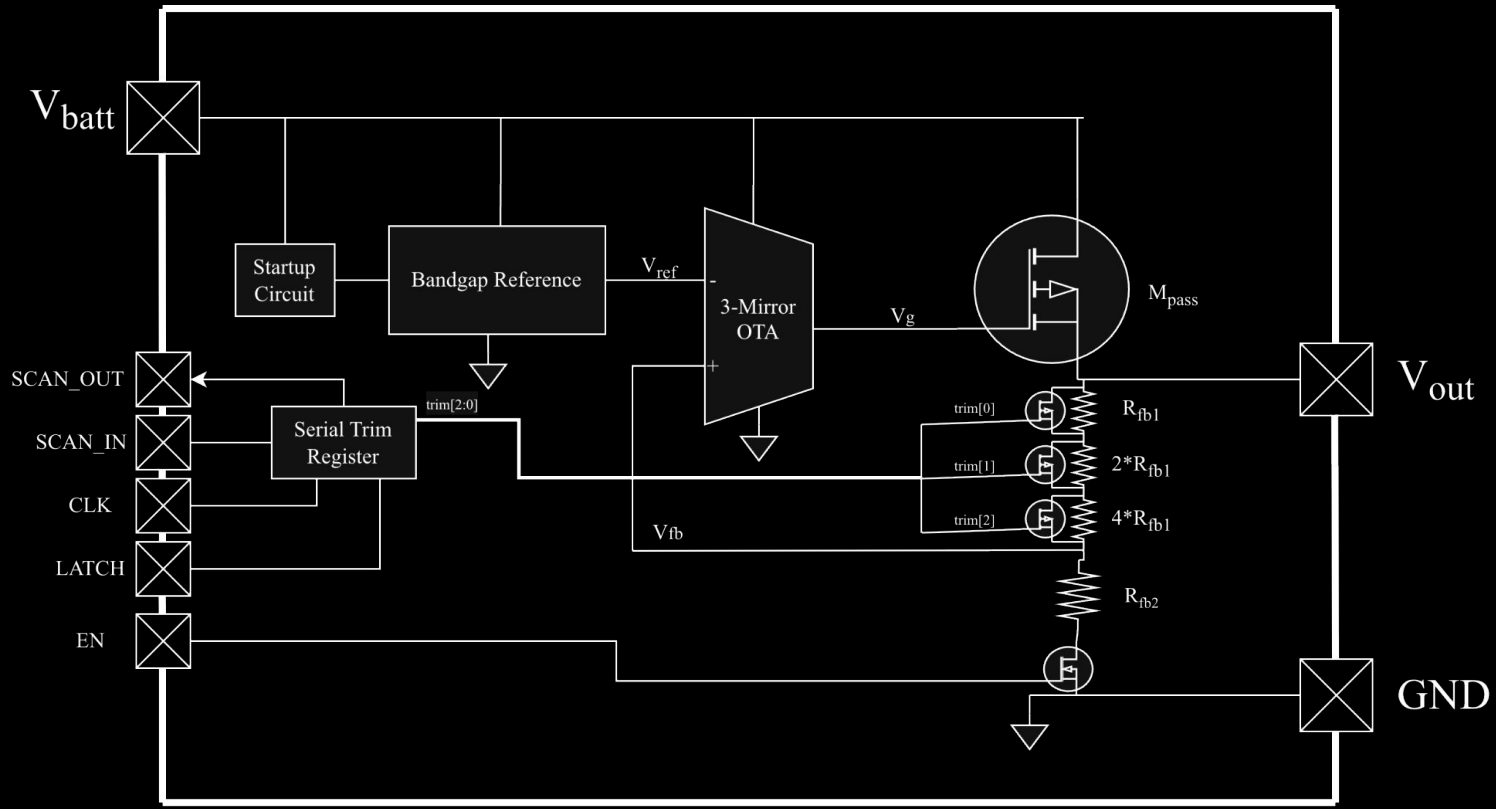
Architecture Overview

Layout Overview

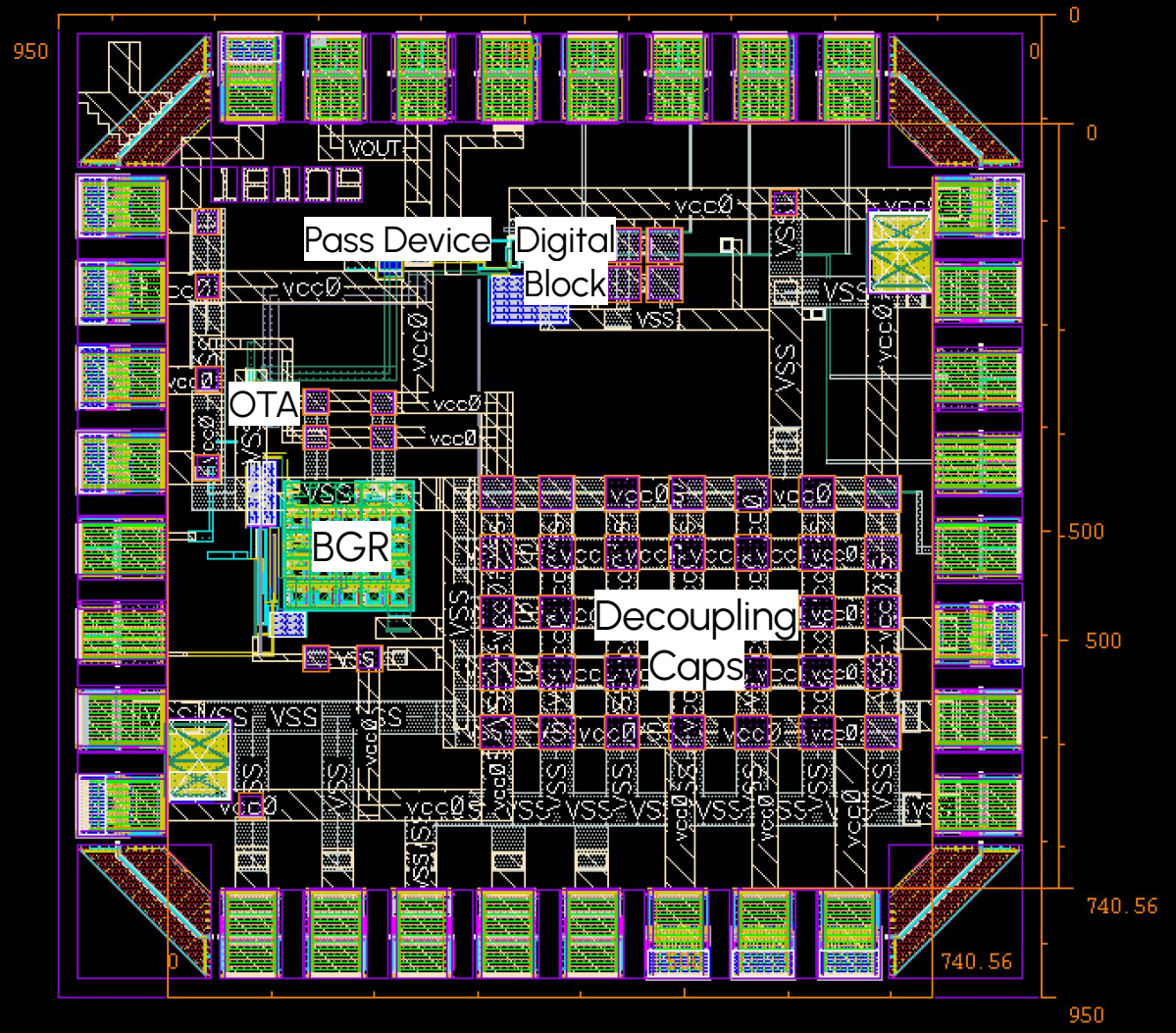
Walkthrough

Outcomes

Architecture Overview



Layout Overview

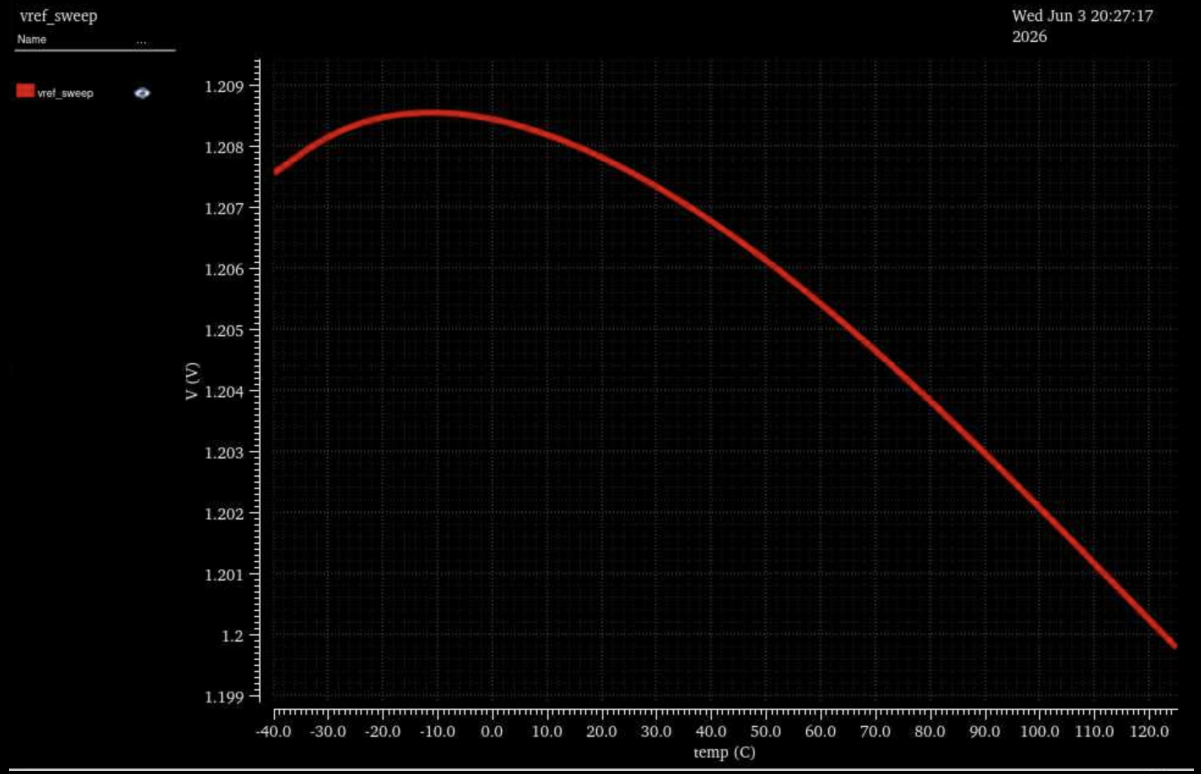


Block Walkthrough

Bandgap Reference

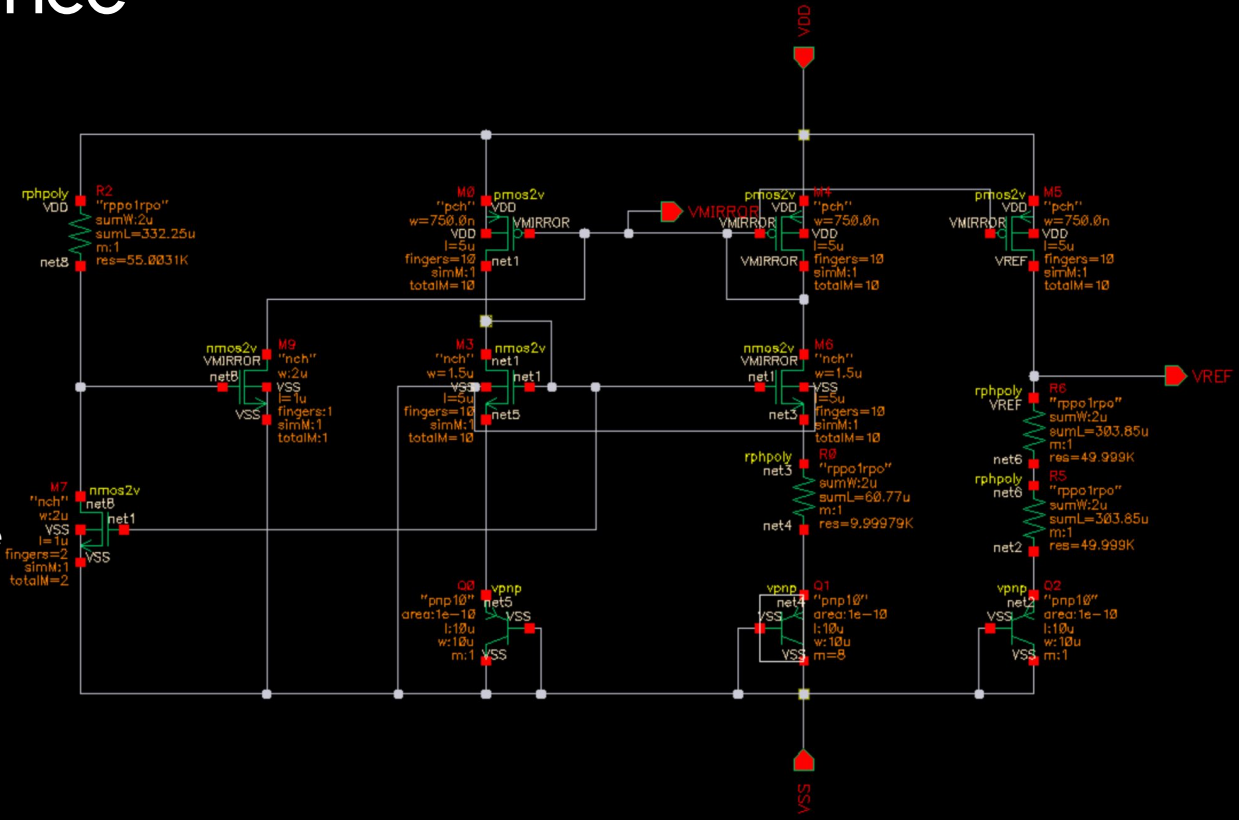
Performance Metric	Value	Performance Metric	Value
V_{REF} @ 27°C	1.207 V	Integrated Noise	4.65 µVRMS
V_{REF} Peak-to-Peak	8.77 mV	Quiescent Current @ 27°C	47.93 µA
Temperature Coefficient	44.02 ppm/°C	Power	47.93 µA * 1.8 V = 86.274 µW
PSRR @ 10 Hz	34.2217 dB	Startup Time	243.8 ms
PSRR @ 1kHz	73.8165 dB	Minimum VDD	1.68 V
PSRR @ 1 MHz	128.019 dB	DC PSRR	105 mV/V
Output Noise @ 1 kHz	2.3585 nV/√Hz	Monte Carlo	.0477%

Bandgap Reference V_{ref} vs. Temperature



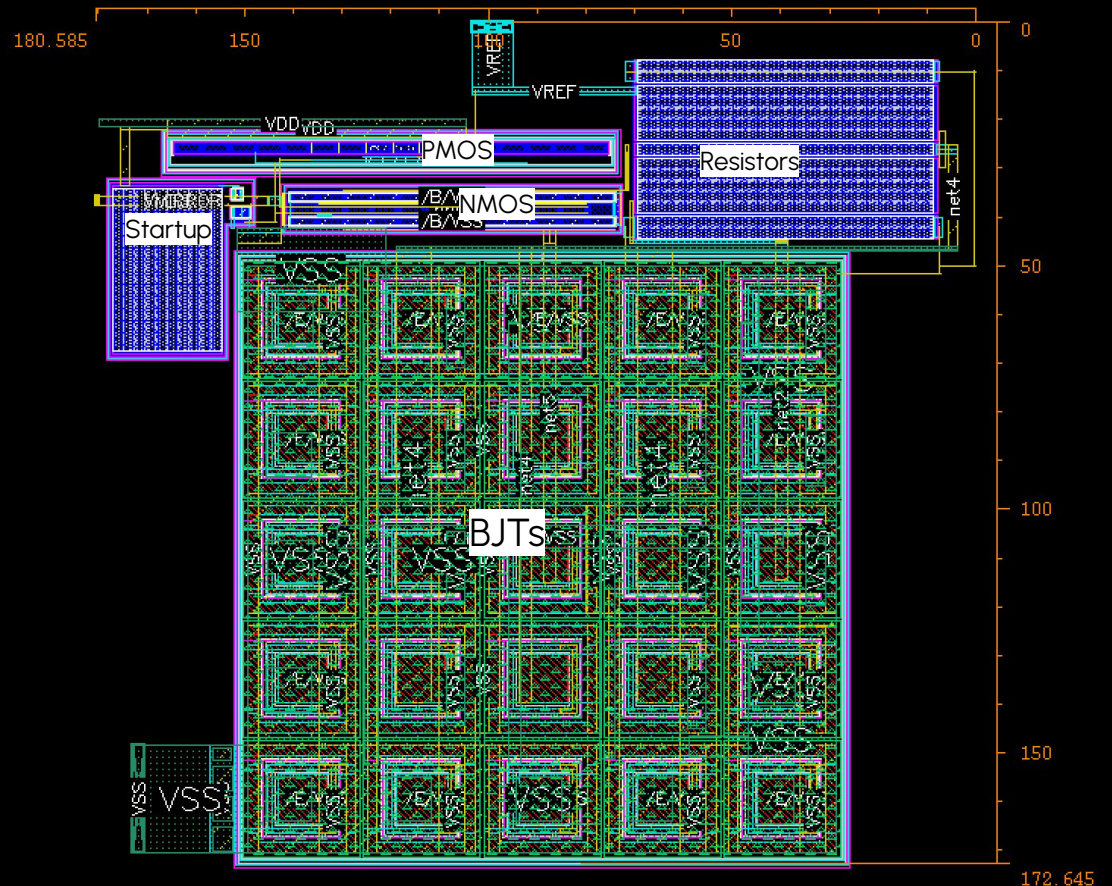
Bandgap Reference

- Current mirror based bandgap reference
- Simulated junction CTAT behavior
  $\partial V_{be}/\partial T$ = -2.021 mV / °C
- Simulated ΔV_{be} PTAT
  $\partial \Delta V_{be}/\partial T$ = 176.1 µV / °C
- Calculated
  R_2 / R = 11.5 → 11
- By equating the current in the two branches
  $I_0 = V_T / R * \ln(n)$
  R ≈ 10 kΩ
  R_2 = 110 kΩ
- Adjust R_2 down to 100 kΩ in order to meet max V_{ref} specs



Bandgap Reference

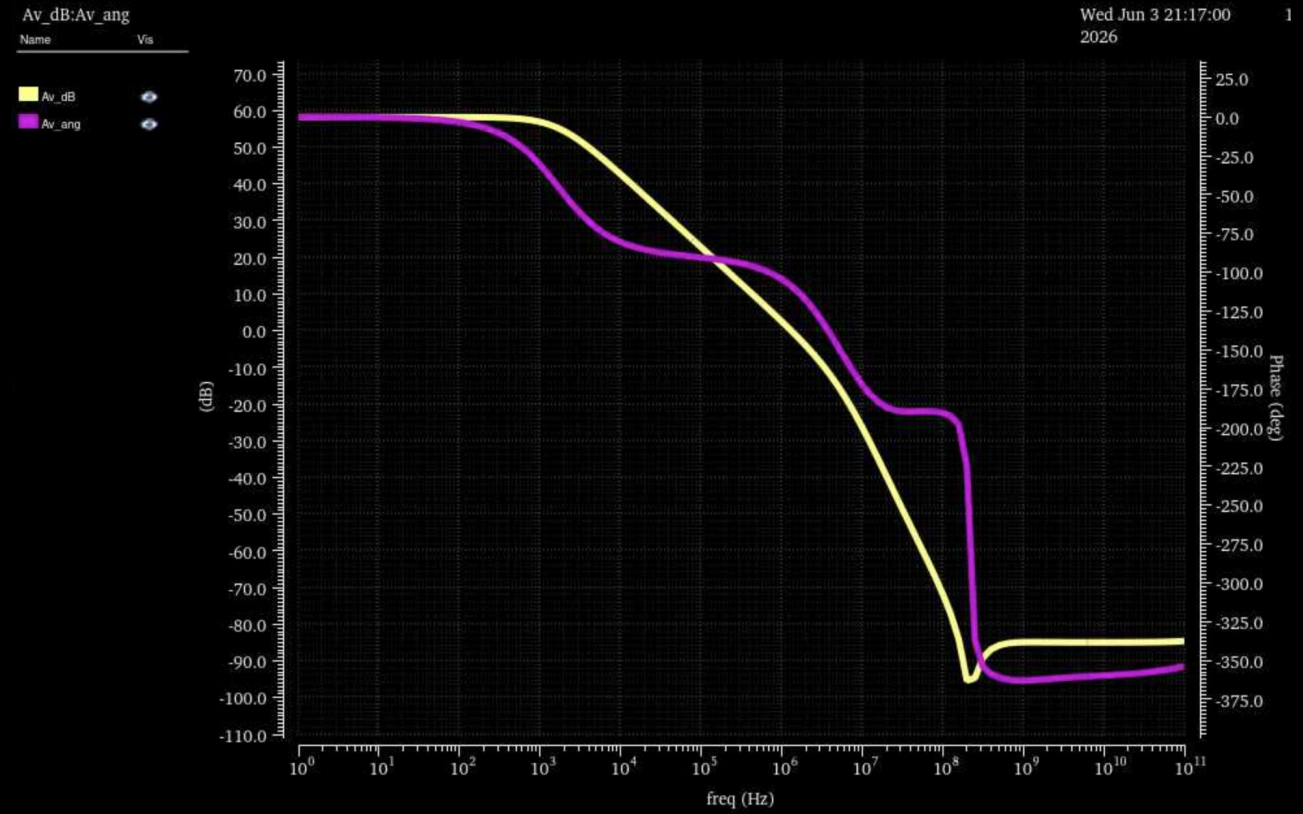
- 180.585 µm x 172.645 µm
- Off-chip noise filter capacitor
- Common centroid layout for BJTs and PMOS current mirrors to reduce effects from linear doping gradients
- Resistor blocks are bordered by dummy resistors
- NMOS fingers that couldn't be interdigitated are instead placed close to each other and laid out in common centroid



OTA

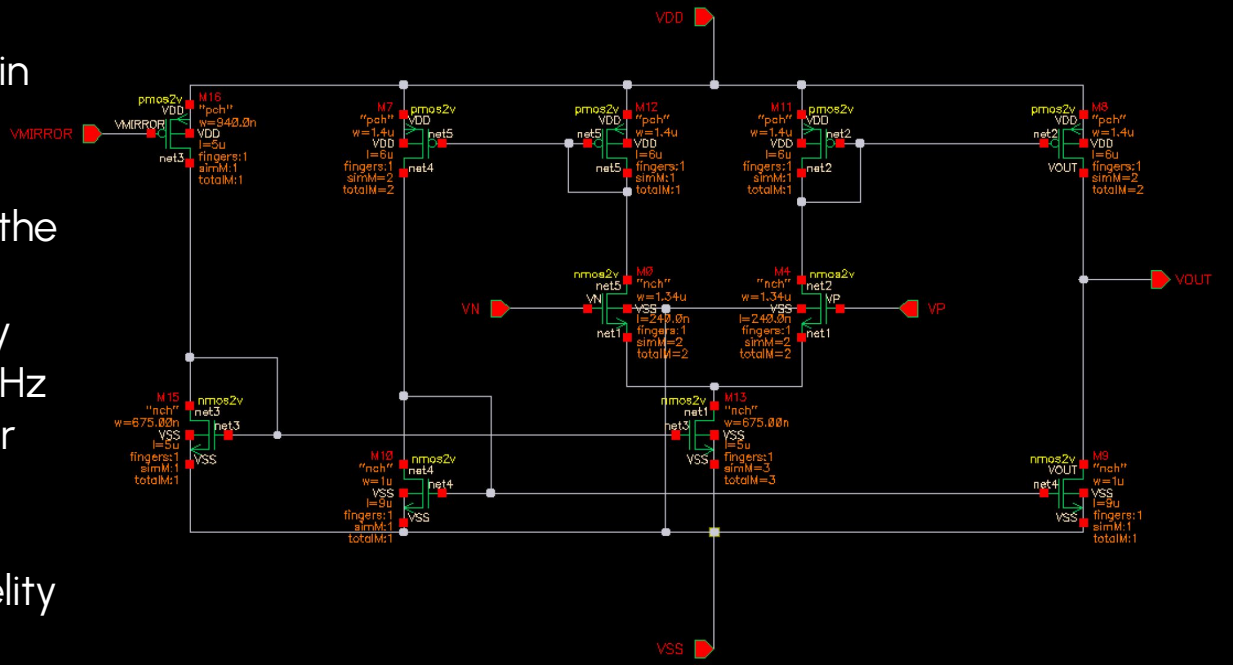
Performance Metric	Value	Performance Metric	Value
DC Gain, AV 0	58.23 dB	PSRR, 10kHz	90.17 dB
Unity Gain Frequency, f0dB	1.32 MHz	Slew Rate	800 mV/µs
Phase Margin (Unity Gain)	71.11 degrees	Input-Referred Noise (DC–1 MHz)	112 μV_{rms}
Power Consumption, PDC	21.89 µW	Output Swing Low	218 mV
CMRR, 10kHz	104.08 dB	Output Swing High	1.615 V

OTA Gain and Phase Plot



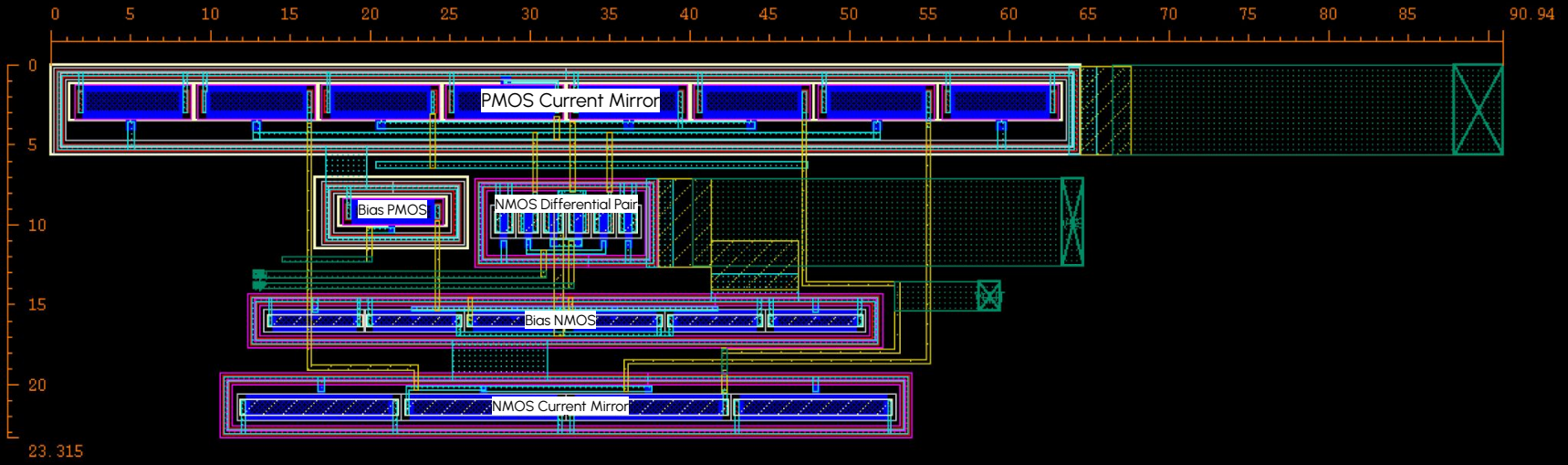
OTA

- 3 mirror topology improves PSRR with a slight increase in gain from the 5T OTA and decouples g_m from r_o by moving the output node to the mirror transistor drains
- Choose g_m = 16 µS to satisfy unity gain frequency of 1 MHz
- Used long devices for mirror transistors and a smaller mirror ratio of 3 to reduce CLM and ensure mirror fidelity



OTA

- 90.94 µm x 23.315 µm
- Matching strategies:
  - Used multipliers for common centroid layout
  - PMOS current mirror pattern: 1-2-3-4-2-1 where 1/2 and 3/4 are pairs
  - Dummies on ends of transistor arrays
- Guard rings around each device/array of devices

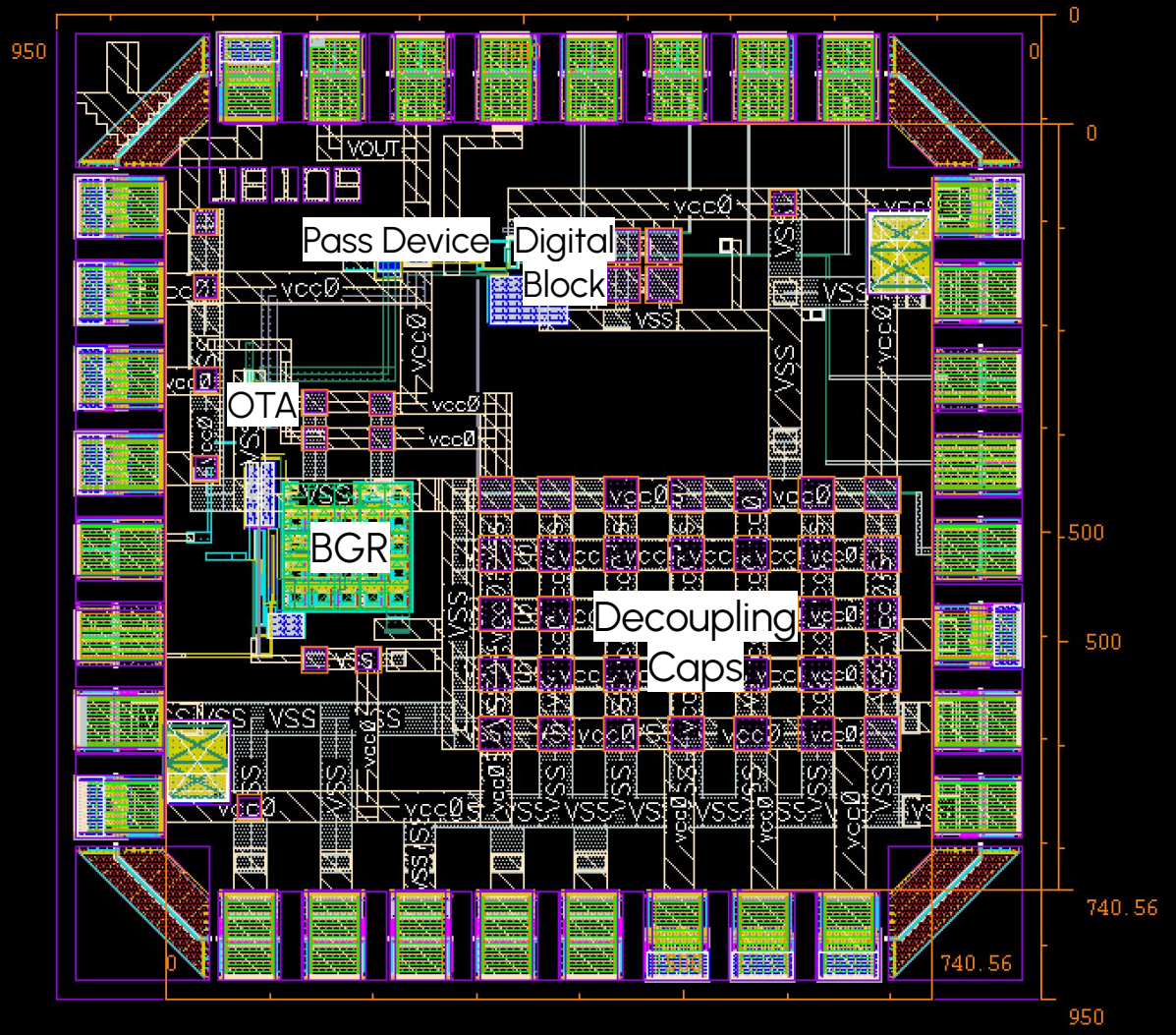


LDO Performance vs. Specification

Performance Metric	Specification Value	Our Value	Performance Metric	Specification Value	Our Value
Output Voltage	1.15 V - 1.25 V	1.208 V	Line Droop	1 mV - 15 mV	366.6 µV
Quiescent Current	50 µA - 100 µA	60.16 µA	Line Overshoot	1 mV - 15 mV	255 µV
Load Regulation	1 mV - 15 mV	3.208 mV	PSRR @ 10kHz	60 dB	60.45 dB
Line Regulation	1 mV - 15 mV	14.96 mV	Phase Margin	min. 45°	90.15°
Load Droop	1 mV - 15 mV	13.97 mV	Total Integrated Output Noise	max. 350 µVrms	78.58u
Load Overshoot	1 mV - 15 mV	9.306 mV	Startup Time	50 µs - 500 ms	247 ms

Top-Level LDO

- Kept BGR, OTA, and pass device near pins
- Used large trunk widths for V_{dd}, V_{ss}, and V_{out}
- Decoupling capacitance on every input power pin, and large decoupling capacitance bank in empty space
- Minimize digital clock trace length
- Compensation network off chip

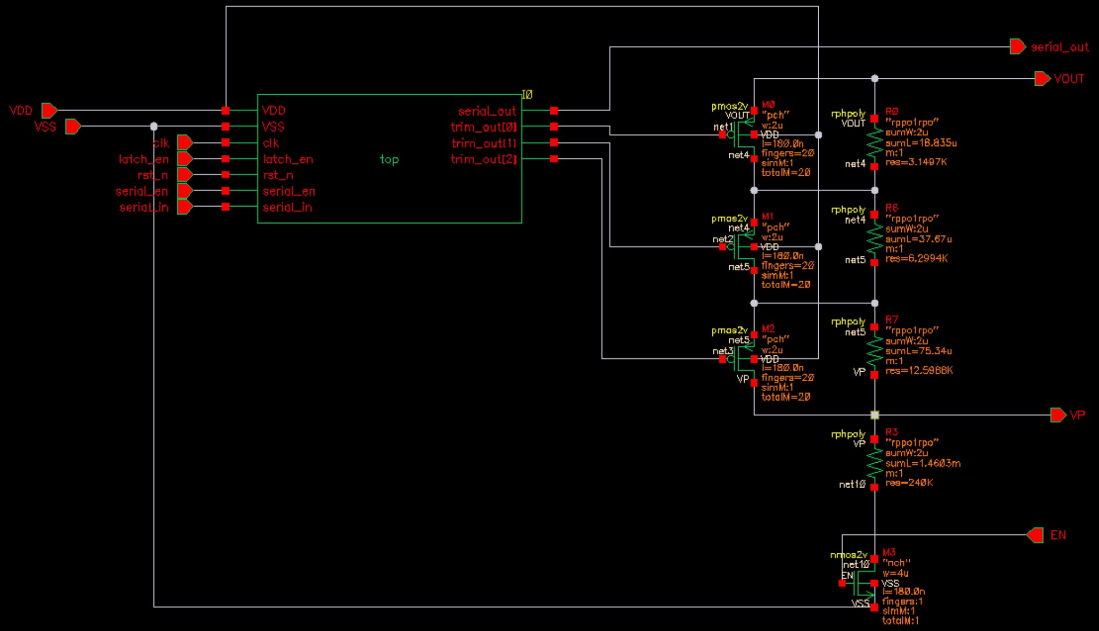


Digital Trim Register

Trim Code	Expected Nominal Vout (V)	Actual Nominal Vout (V)
Enable OFF	1.209	1.208
000	1.21	1.209
001	1.225	1.225
010	1.241	1.241
011	1.257	1.256
100	1.273	1.272
101	1.288	1.288
110	1.304	1.304
111	1.319	1.319

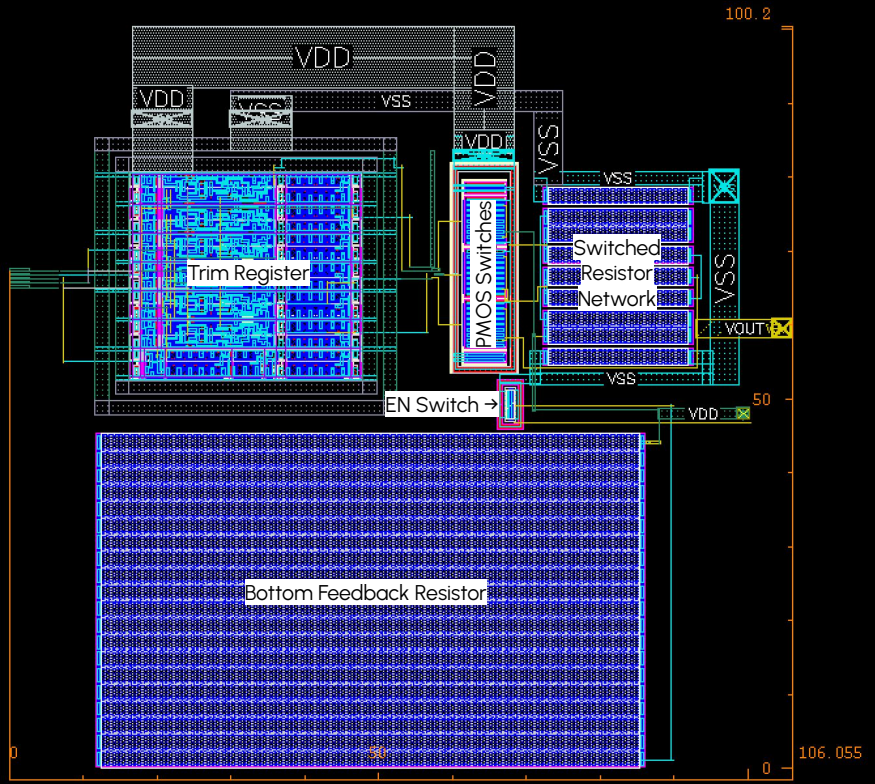
Digital Trim Register

- Trim out controls 3 PMOS that bypass each resistor
- Resistors are configured as R, 2R, and 4R for linearity of voltage change ($\Delta V \approx 15$ mV)
- FB network and digital block increase I_Q by 5.03 µA



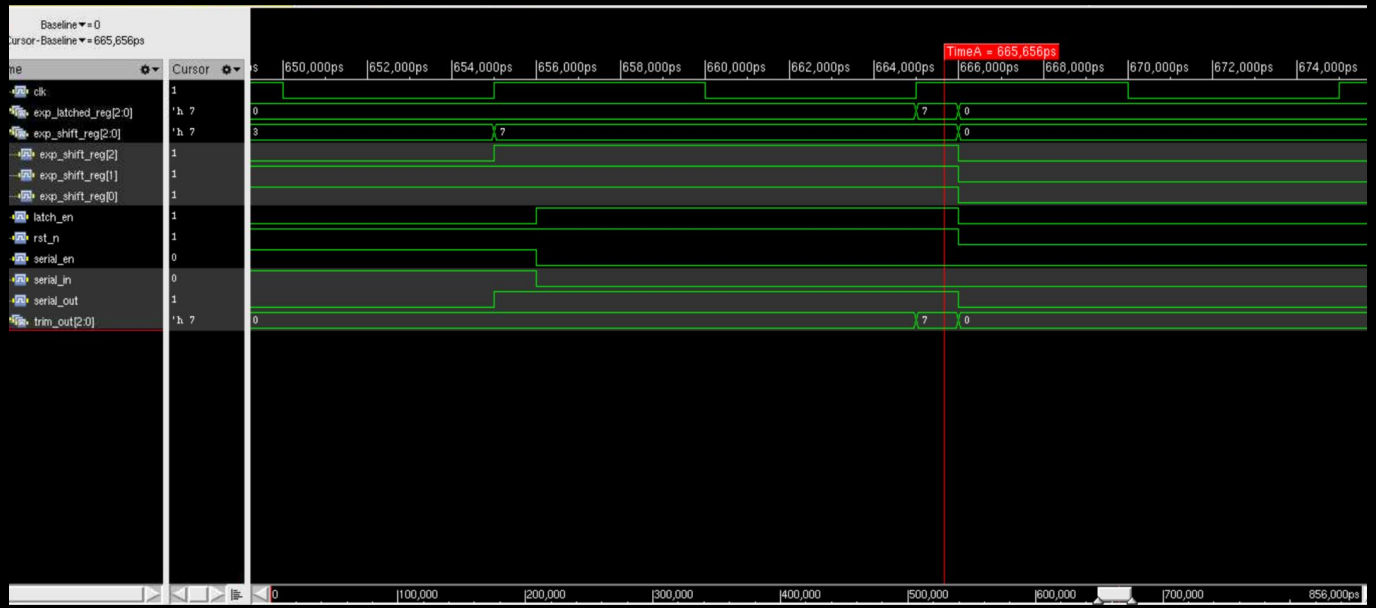
Digital Trim Register

- Common centroid layout for switched resistor network
- Dummies on ends of PMOS array



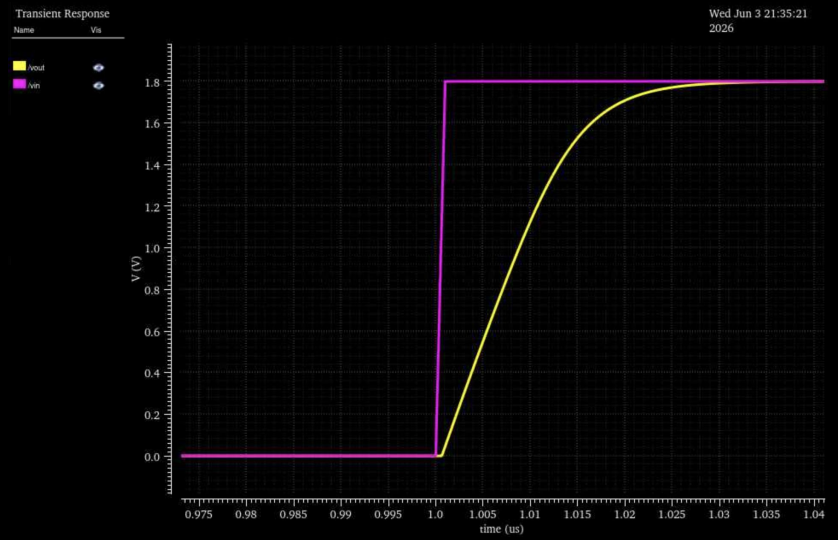
Digital Trim Register – Testing

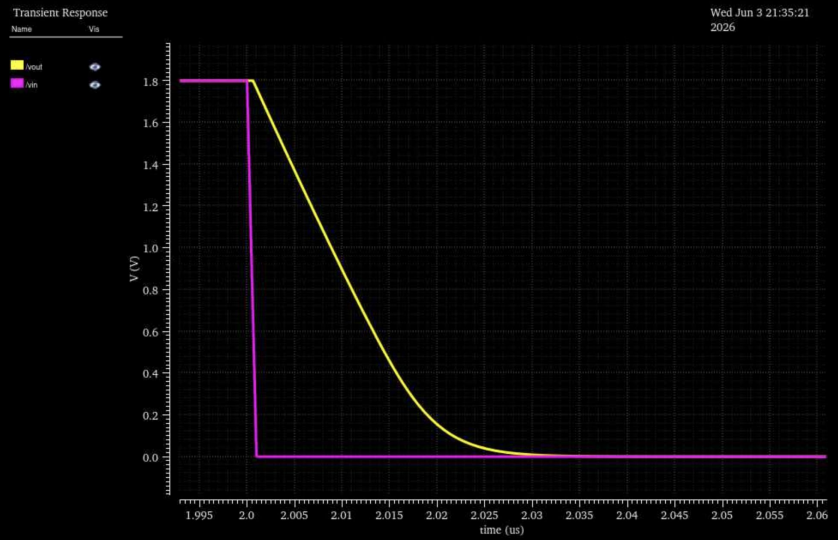
- Ran gate-level simulation on the post-layout netlist
- Verified waveforms against expected behavior
- Ensured that trim register held the correct values after a positive clk edge when latch_en was held high



Digital Output Buffer

- Designed a minimum size output buffer made of two inverters for the scan_out output pin
- Tested with PEX extracted layout to drive a 10 pF load through the bondwire
- 10% to 90% rise time is <20ns





Outcomes

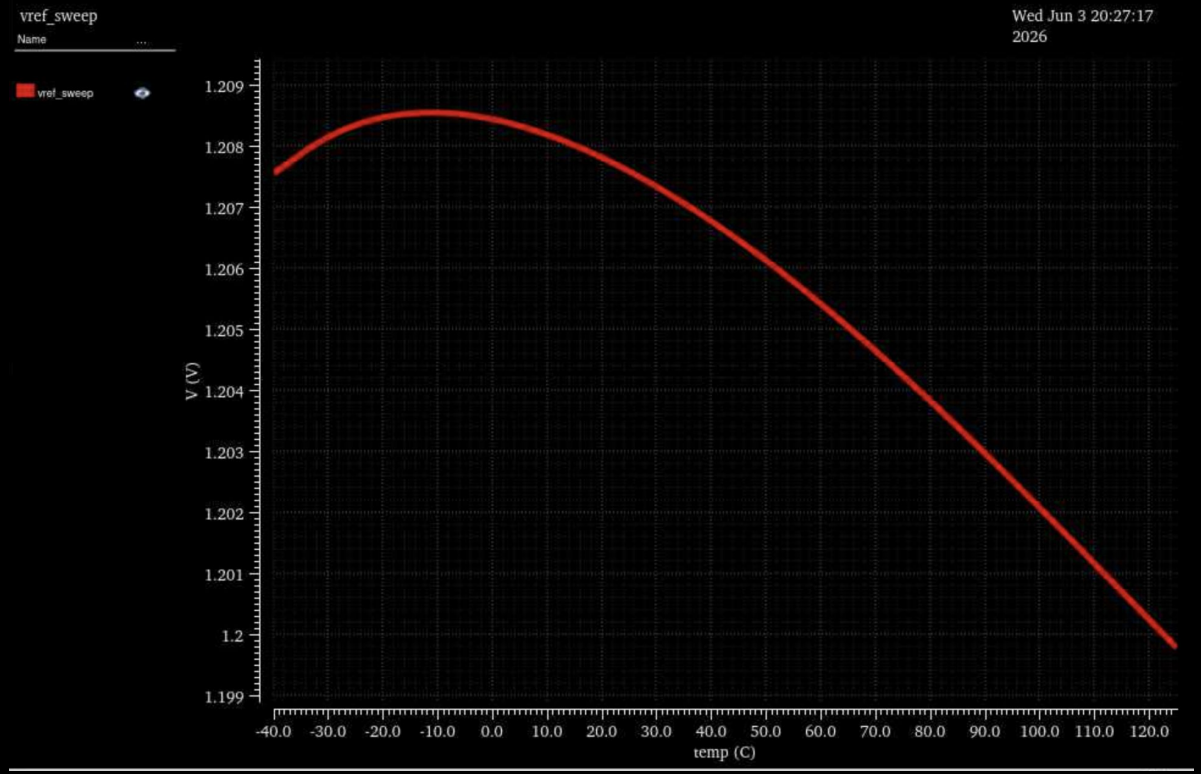
- Successes
  - PEX simulation results consistently met deviation specs
  - Digital trim range nominal Vout values are within ~1mV of expected values
- Challenges
  - Digital block integration introduced many DRC errors that were difficult to solve
  - Lack of mixed-signal LVS due to tooling issues meant extreme care was needed during integration
- Improvements for next time
  - Consider top-level when laying out individual blocks – current mirrors across blocks are not interdigitated
  - Simulate early and often. PEX simulations were very time consuming!

Thank You!

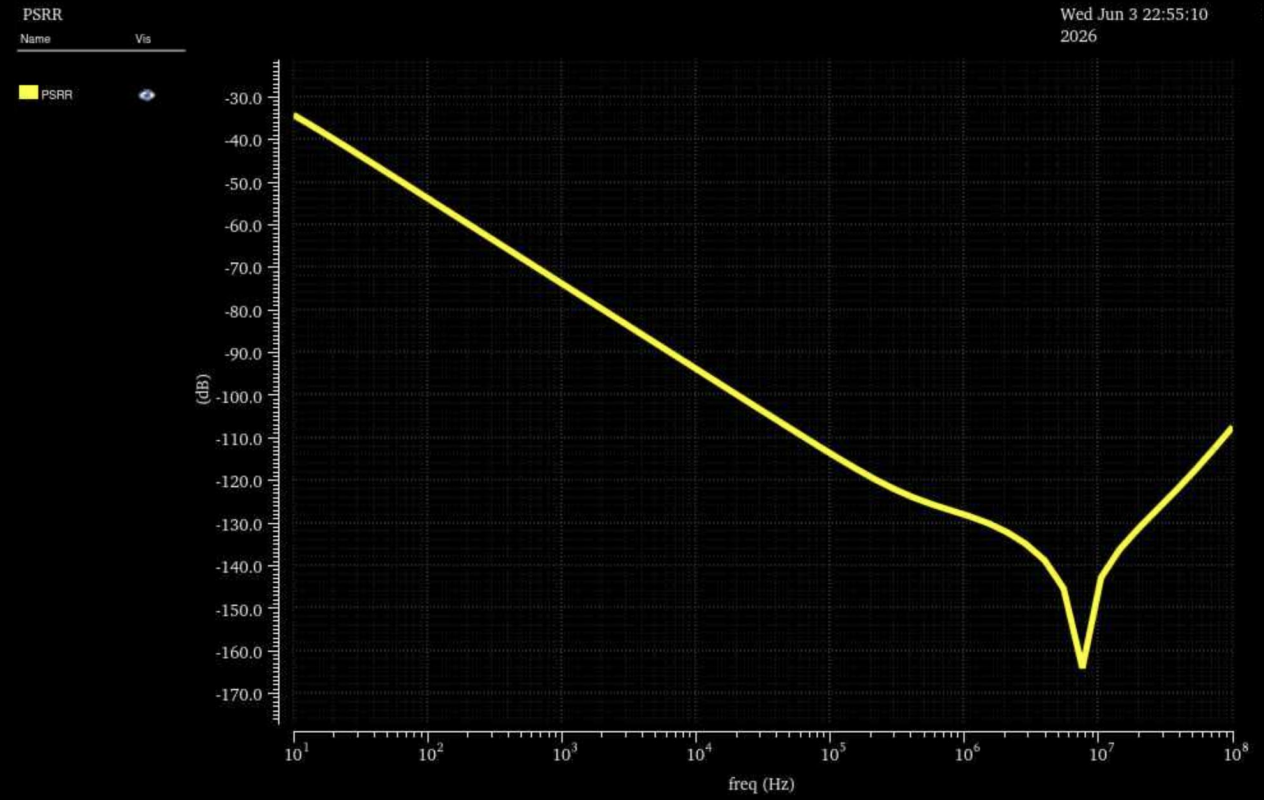
Questions?

Appendix: Graphs

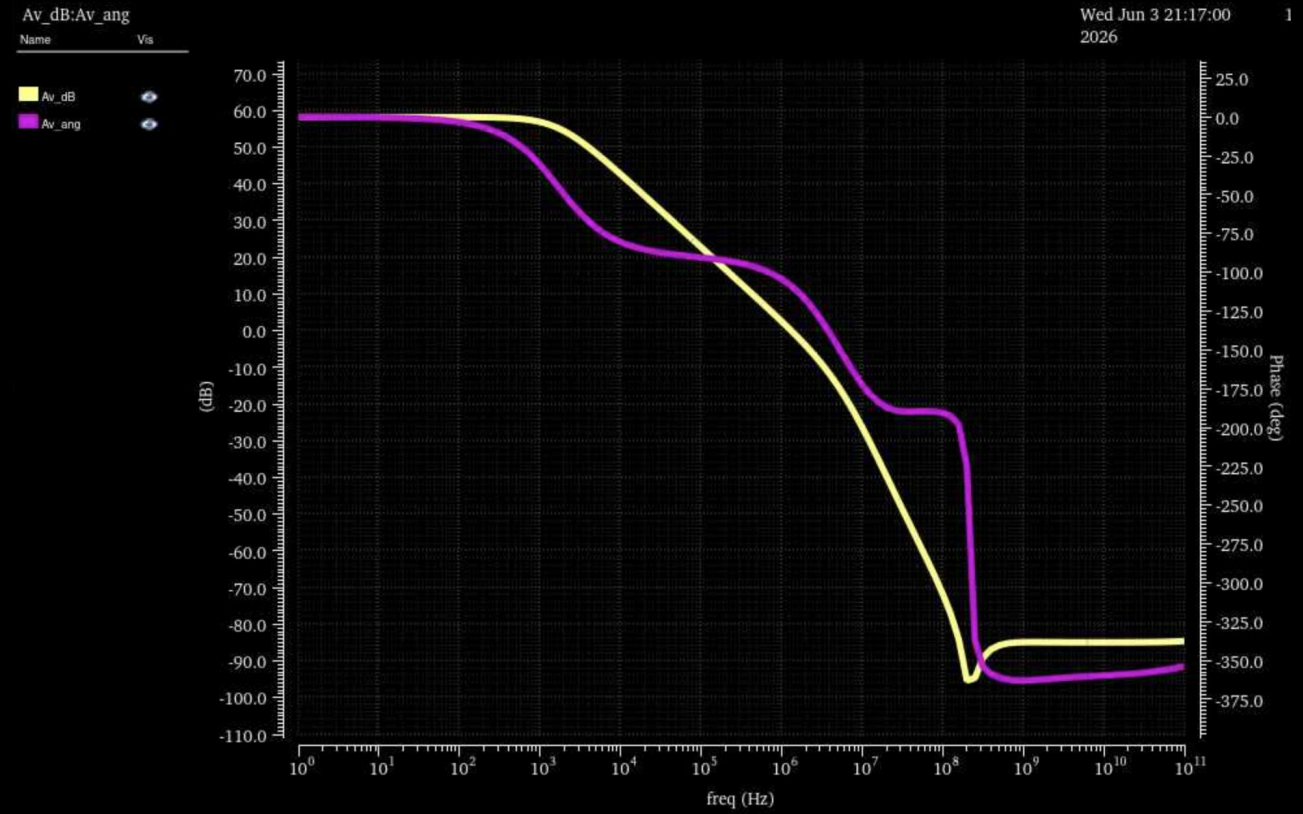
Bandgap Reference V_{ref} vs. Temperature



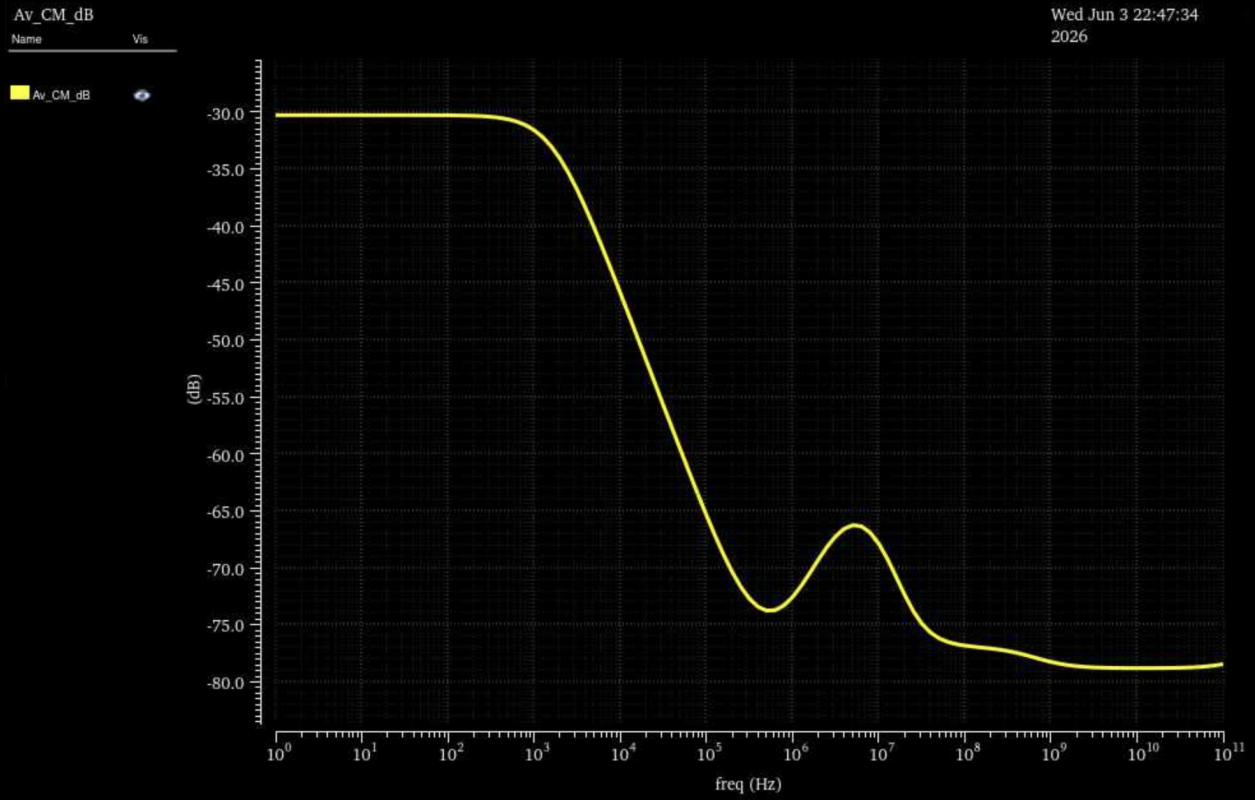
BGR PSRR


PSRR
Name
Vis
PSRR
Wed Jun 3 22:55:10
2026
-30.0
-40.0
-50.0
-60.0
-70.0
-80.0
-90.0
-100.0
-110.0
-120.0
-130.0
-140.0
-150.0
-160.0
-170.0
(dB)
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
freq (Hz)

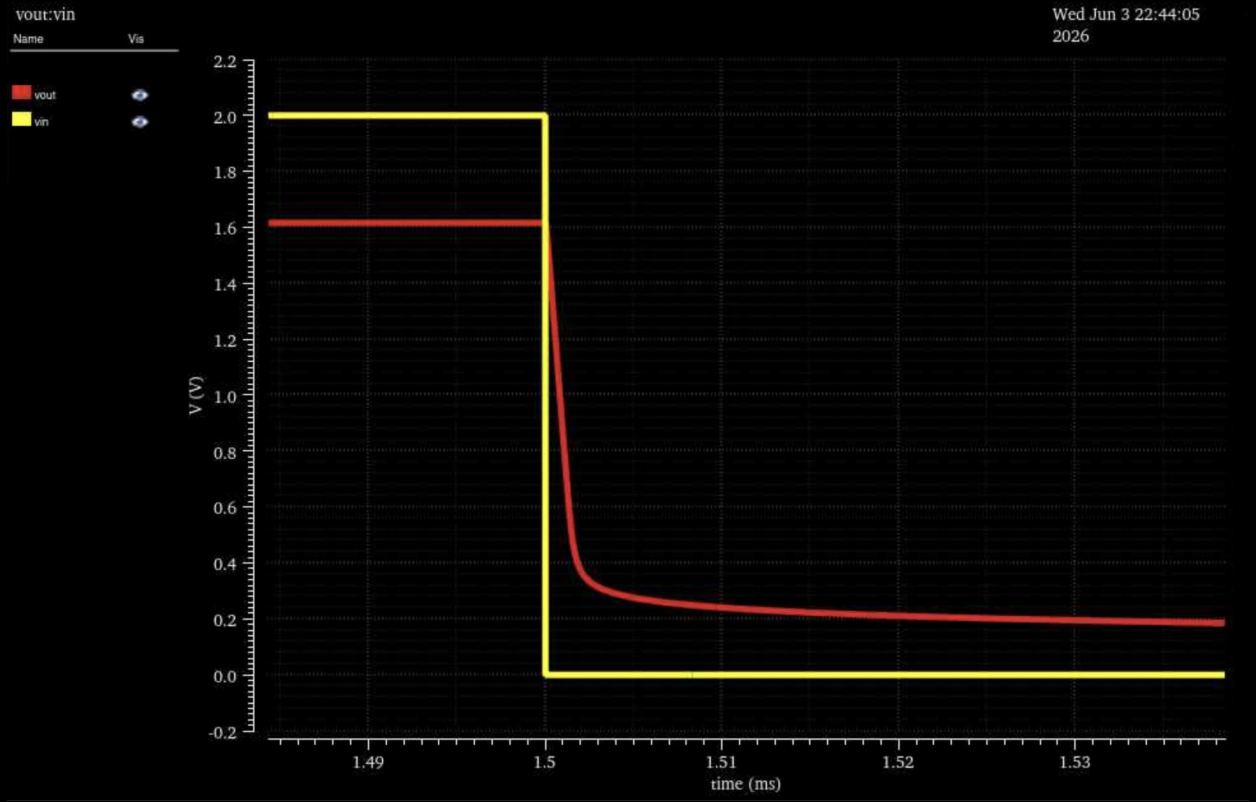
OTA Gain and Phase Plot



OTA Av CM


Av_CM_dB
Name
Vis
Av_CM_dB
Wed Jun 3 22:47:34
2026
-30.0
-35.0
-40.0
-45.0
-50.0
-55.0
-60.0
-65.0
-70.0
-75.0
-80.0
(dB)
10^0 10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9 10^{10} 10^{11}
freq (Hz)

OTA Slew Rate



Top-level LDO Load Transient Response

